Part A – Theory & Pre-implementation

1) RC4 Overview

RC4 is a symmetric key stream cipher. It generates a pseudo-random keystream which is XORed with the plaintext to produce ciphertext.
• Type of cipher: Stream cipher
• Role of key: Used to initialize and shuffle the state array
• Keystream: A random byte sequence generated using PRGA
• Why symmetric: The same key is used for both encryption and decryption

Encryption and decryption:
Ciphertext = Plaintext ⊕ Keystream
Plaintext = Ciphertext ⊕ Keystream

2) Key Scheduling Algorithm (KSA)

1. Initialize state array S of size 256 → S[i] = i
2. Repeat key using key [i % keylen]
3. j = (j + S[i] + key [i % keylen]) mod 256
4. Swap S[i] and S[j]

Modulo 256 keeps the index within range.

3) Pseudo Random Generation Algorithm (PRGA)

1. Initialize i = 0, j = 0
2. i = (i + 1) mod 256
3. j = (j + S[i]) mod 256
4. Swap S[i] and S[j]
5. t = (S[i] + S[j]) mod 256
6. Keystream byte = S[t]
7. XOR with data

4) Encryption & Decryption Process

RC4 uses XOR operation.

Plaintext = A (65)
Keystream = K (75)
Ciphertext = 65 ⊕ 75
Decryption: Ciphertext ⊕ 75 = 65 = A

Same algorithm is used because P ⊕ K ⊕ K = P.

Part B – Implementation

Functions implemented:
1. ksa(key, keylen, S)
2. prga(S, data, len)
3. rc4(key, keylen, data, len)
4. main() – accepts key and plaintext, performs encryption and decryption.

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Sample Input

Enter key: secret
Enter plaintext: hello

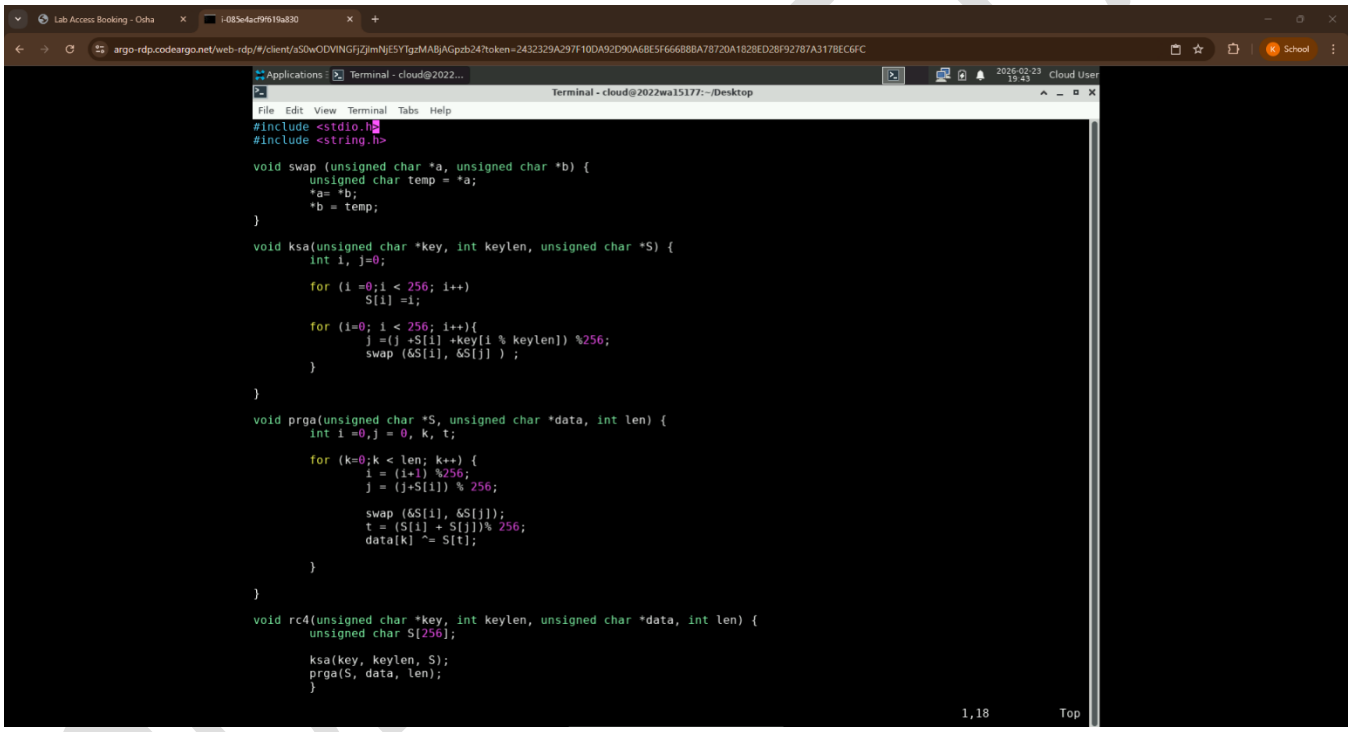
Sample Output

Ciphertext (Hex): XX XX XX XX XX
Decrypted text: hello

Result

The RC4 algorithm was successfully implemented. Encryption and decryption were performed using the same key and the original plaintext was retrieved.

Screenshot of program written in vi editor.



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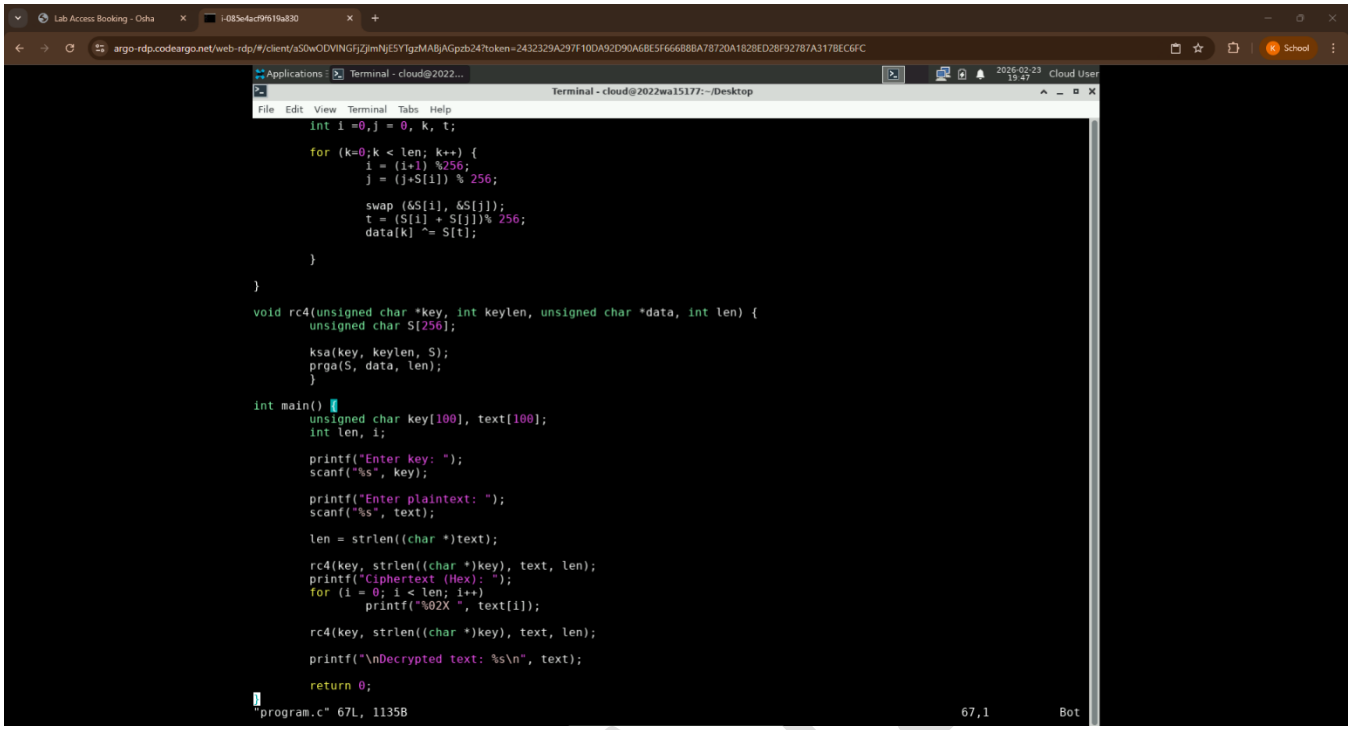
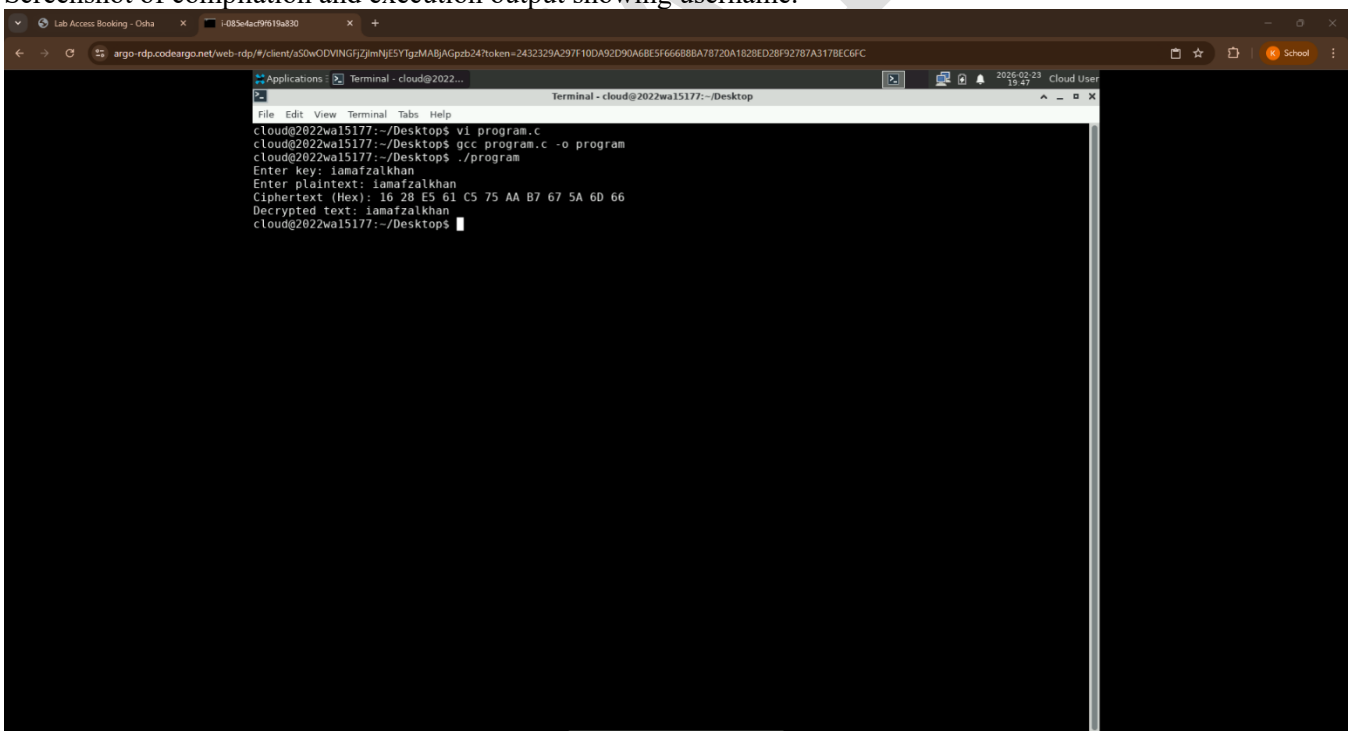


Screenshot of compilation and execution output showing username.



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Observation

• RC4 is fast due to simple operations.

• Same function is used for encryption and decryption.

• XOR operation makes the algorithm reversible.

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